Course Syllabus

General Information

Faculty	Information Technology		
Department	Computer Science		
Semester	Second	**Academic Year**	2025/2026

Course Details

Course Title	Internet of Things			**Course No.**	A0413404		
Type of Learning	■ On- campus ☐ Blended (……..….. Platform) ☐ Online (……..….. Platform)						
Credit Hours	3	**Theoretical**	3	**Practical**	0		
Course Level (according to JNQF)			7	**JNQF Hours***	16.2		
Pre-requisite	Computer and Network Security -0413405			**Co-requisite**			

Course Instructor Information

Instructor Name	Dr. Rashed Alazzeh		
Instructor E-mail	razzeh@meu.edu.jo		
Course Time	(11:00-12:00 Sun, Tue, Thr),		
Office Hours	10:00 – 11:00 (Sun, Tue)		
Office Number	N108	**Office Phone**	
Lab Instructor Name –If assigned-			

Sources and References

1. Textbook

IoT fundamentals: Networking technologies, protocols, and use cases for the internet of things, Hanes, David, CISCO, 2017

2. References

- Hands-On Internet of Things with Blynk: Build on the power of Blynk to configure smart devices and build exciting IoT projects, Seneviratne,2018
- THE INTERNET OF THINGS (IoT) PRACTICAL HANDBOOK: IoT Practical Handbook, Idris et al, 2021
- Schwartz, Marco. Internet of things with arduino cookbook. Packt Publishing Ltd, 2016.-
- The Internet of Things: Enabling Technologies, Platforms, and Use Cases", by Pethuru Raj and Anupama C. Raman (CRC Press), 2017
- Internet of Things: A Hands-on Approach", by Arshdeep Bahga and Vijay Madisetti (Universities Press) 2015.

Teaching Methods

1. Lecture-based learning
2. Technology-based learning
3. Group learning
4. Individual learning
5. Inquiry-based learning
6. Expeditionary learning

Course Description

This course provides a foundation in the Internet of Things, including: the architectures, sensors, actuators, components, tools, and analysis by teaching the concepts behind the IoT and a look at real-world solutions and applications along with security and privacy issues.

Course Learning Outcomes (CLOs) and its correlation with Program Learning Outcomes (PLOs) and JNQF Descriptors

No.	CLOs	PLOs	JNQF Descriptors		
			Knowledge	Skill	Competency
1	Define the IoT, its benefits, and applications.	PLO1	✓		
2	Describe the IOT Architecture and its components	PLO1	✓		
3	Explain the infrastructure for supporting IoT deployments	PLO1	✓		
4	Discuss IoT protocols for communication and related issues including security and privacy.	PLO2		✓	
5	Design and Implement IoT applications to solve problems using Microcontrollers	PLO6		✓	

Course Topics

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
1	**Course Syllabus Discussion & Introduction to the course**	Lecture (1)	L	**1**	-	F
		Lecture (2)	L	**1**	-	F
2	**What is IoT Applications of IoT**	Lecture (1)	L	**1**	1,3,4	S
		Lecture (2)	L	**1**	1,3,4	S
3	**IoT Network Architecture and Design**	Lecture (1)	L	**1**	1,3	S
		Lecture (2)	L	**2**	1,3	S
4	**IoT Network Architecture and Design**	Lecture (1)	L	**2**	1,3	S
		Lecture (2)	P	**3**	1,3	S
5	**Smart Objects The "Things" in IoT**	Lecture (1)	L	**2**	1,3,4	S
		Lecture (2)	P	**3**	1,3,4	S
6	**Connecting Smart Objects + Blynk**	Lecture (1)	L	**3**	1,3,4	S
		Lecture (2)	P	**5**	1,3,4	S
7	**IP as the IoT Network Layer**	Lecture (1)	L	**3**	1,3,4	
		Lecture (2)	P	**4**	1,3,4	S
8	**Projects made with Blynk Mid-Term**	Lecture (1)	P	**2**	4	S
		Lecture (2)	L	**-**	-	S

Week	Topic	Lecture Type		CLOs	Assessment	
		Lecture No.	Lecture Type		Assessment Tool	Summative/ Formative
9	**Microcontrollers introduction**	Lecture (1)	L	**5**	2,3,4	S
		Lecture (2)	P	**5**	2,3,4	S
10	**Microcontroller Design: Arduino, Raspberry Pi**	Lecture (1)	P	**5**	2,3,4	S
		Lecture (2)	P	**5**	2,3,4	S
11	**Sensors, Actuators**	Lecture (1)	L	**2, 3, 4,5**	2, 4	S
		Lecture (2)	P	**2, 3, 4,5**	2, 4	S
12	**IoT Security: Application Protocols for IoT and security issues**	Lecture (1)	L	**4**	2, 4	S
		Lecture (2)	P	**4**	2, 4	S
13	**Data and Analytics for IoT**	Lecture (1)	L	**4**	2, 4	S
		Lecture (2)	P	**4**	2, 4	S
14	**IoT and Cloud Computing**	Lecture (1)	L	**2**	2, 4	S
		Lecture (2)	P	**4**	2, 4	F
15	**Project Presentation and Discussion** **Final Exam**	Lecture (1)	P	**1,2,3,4,5**	4	F
		Lecture (2)	P	**-**		

Course Evaluation Time and Weight

Assessment Tool	Weight %	Expected Due Date
1 Mid Written Exam.	25	W8
2 Final Written Exam.	40	W16
3 Project	35	W14,W15

Course Policies

Item	Policy
Quizzes	
Exams	▪ The format of the exams is generally (but NOT always) as follows: General Definitions, Multiple-Choice, True/False, Analyze a Problem, Essay Questions, etc.
Makeup Exams	▪ Makeup exam should not be given unless there is a valid excuse.
Drop Date	▪ Last day to drop the course is according to the University regulations.
Academic Honesty	▪ Cheating or copying on exam or quiz is an illegal and unethical activity. ▪ Standard MEU policy will be applied. ▪ All graded assignments must be your own work (your own words).
Attendance	▪ Excellent attendance is expected. ▪ MEU policy requires the faculty member to assign ZERO grade if a student misses 10% of the classes. ▪ Sign-in sheets will be circulated. ▪ If you miss class, it is your responsibility to find out about any announcements or assignments you may have missed.
E-Learning	▪ It is your responsibility to check the course's eLearning website regularly (at least once every 2-3 days) for announcements and material.
Workload	▪ The student should expect to spend 6 hours per week as an average workload.
Graded Exams	▪ Instructor should return exam papers graded to students within one week after the exam date.
Participation	▪ Participation in and contribution to class discussions will affect your final grade positively. Raise your hand if you have any question. ▪ Making any kind of disruption and (side talks) in the class will affect you negatively.
Assignment/ Projects	▪ Homework's are as mentioned in the previous table. ▪ Students organize themselves in teams of (2-3) students each. Each team will give at least one presentation regarding a selected case, delivering a fully working application. ▪ More details about phases and topics of the projects will be given later on.
Others	

Rubrics and Feedbacks (Project)

Criteria	Excellent	Very Good	Adequate	Poor
Creativity (Approach and Topic):	shows deep insight, and significant understanding of the problem; goes beyond the brief and demonstrates significant originality in the ideas and their application; uses precedents and/or materials in unexpected ways; surprising and delightful outcome. (5 marks)	shows engagement, exploration and insight; uses precedents and/or materials in relatively original ways. (4 marks)	satisfies minimum requirements of the brief and/or a minor increment over existing work (extension/adaptation of a precedent) (2 marks)	does not satisfy brief and no originality or creativity demonstrated (e.g. direct replication of prior approach without extension) (1 mark)
Context (Concept and Background):	provides a thoughtful and considered introduction to the work and supports the context with relevant references and critical analysis of past work.(6 marks)	provides a thoughtful and considered introduction to the work with limited references and limited analysis of past work. (4 marks)	satisfies minimum requirements. (2 marks)	does not satisfy brief (no mention of problem space, scenario) and/or does not include precedents. (1 marks)
Execution	provides a strong technical understanding; well considered component choices; functions and is well organized; and works with advanced components, implements functionality beyond the brief and demonstrates technical skill , shows competence in the design and material choices (8 marks)	functional; demonstrates understanding of components and applied them in a working circuit; circuit is well organized; component selection is sensible and demonstrates some technical skill. that form is sensible and appropriate. (5 marks)	satisfies minimum requirements - provides the minimum core functionality, is basic in operation and could be improved. (3 marks)	does not satisfy brief - code & circuit not included or does not compile. (1 mark)
Communication (Documentation, Presentation):	as good, but includes an additional level of rigor, reflection or professionalism that elevates the outcome. e.g. Documentation that is ready to share online for press or crowdfunding. (6 marks)	clearly communicated and sufficiently detailed to easily repeat the outcome; well and appropriately illustrated with media, diagrams and code. (4 marks)	satisfies minimum requirements - it contains the required components; but reasonably poor quality, verbose, unclear or shows other communication issues. (2 marks)	documentation is missing or doesn't provide any illustration (1 mark)

Rubric detail for final IoT project

	Missing Information	**Needs Improvement**	**Meets Expectations**
Problem Definition & Use Case	Points: 0 (0.00%)	Points: 0.5 (10%)	Points: 1 (20%)
Hardware Componenets	Points: 0 (0.00%)	Points: 0.5 (10%)	Points: 1 (20%)
Communication Protocols	Points: 0 (0.00%)	Points: 0.5 (10%)	Points: 1 (20%)
Cloud/ Backend Platform	Points: 0 (0.00%)	Points: 0.5 (10%)	Points: 1 (20%)
Documentation & Presentation	Points: 0 (0.00%)	Points: 0.5 (10%)	Points: 1 (20%)
			Total Marks: 35
			$Total\ Marks = \left(\sum_{i=1}^{5} Ci\right) x\ 5$
			All relevant laboratory experiments are assigned 10 marks

* **JNQF Hours Calculation**

Activity	Duration/ week	Times/ Semester	No. of Hours
Learning Activities			
Lectures and Seminars	2	15	30
Tutorials	0	0	0
laboratory	2	15	30
Activities for Assessment and Evaluation			
First Exam/ Midterm	1	1	1
Second Exam (If applicable)			
Final Exam	2	1	2
Quizzes	0	0	0
Self-Learning Activities			
Assignments/ Homework	4	4	16
Case Studies	4	2	8
Presentations	1	1	3
E-learning			
Extra Readings	6	12	72
Total Hours	**162**		
JNQF Hours	**Total Hours/10 = 16.2**		